

Public Sentiment and Common Topics on Philippine Agriculture: A Comparative Text Analysis of Reddit Discussions and Online News Articles

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Abstract—This study explores public sentiment and common topics related to agriculture in the Philippines using data from Rappler articles and Reddit posts. Basic analysis, topic modeling, and sentiment classification show that both sources highlight similar challenges faced by farmers. Rappler articles remain mostly neutral, while Reddit posts express more negative reactions to agricultural problems. Overall, the findings show that public discussion and news reporting reflect the same major concerns about the country's agricultural sector.

I. INTRODUCTION

A. Background and Brief Related Literature

The main intuition in this study is to provide context into what are the common topics and sentiments towards the topic **agriculture** in the Philippines. While it is common knowledge that agriculture is often **unappreciated** and that it is **unsupported**, this study provides a textual analysis on the conversations that occur in online platforms.

The Department of Agriculture (DA) recognizes the fact that the agricultural situation in the Philippines is struggling [1]. These struggles are stem from various challenges that the sector is facing, such as the decreasing number of farmers, harvestable area, and the increasing cost of production [2]. With the continuous rise of population, the demand for food is also increasing. However, the agricultural sector is not able to keep up with this demand due to various challenges such as lacking of government support, climate change, and limited access to new farming technology [2].

Considering the challenges that the agricultural sector is facing, it is important to understand the public sentiments and common topics associated with agriculture in the Philippines. This study aims to analyze the public sentiments and common topics associated with agriculture in the Philippines by utilizing text data from Reddit discussions and rappler news articles. By comparing the sentiments and topics from these two, we will be able to identify the differences and similarities in the public discourse on agriculture in the Philippines.

The findings of this study can provide insights into the public perception of agriculture in the Philippines. Albeit

the dataset is rather sparse or limited , it can still provide significant insights into the public discourse on agriculture in the Philippines. Comparing Reddit discussions and Rappler articles, it is possible to identify the gaps and needs of agricultural sector in the country by understanding if the news articles are addressing the concerns of the public, commonly expressed in the widely open Reddit platform.

B. Research Questions

- 1) What are the dominant public sentiments toward agriculture in the Philippines across Reddit discussions and news articles?
- 2) What are the most common topics associated with Philippine agriculture in public (Reddit) versus formal (news) discourse?
- 3) How do public sentiments and topics differ between the two sources?
- 4) Is it possible to identify the specific needs or concerns of farmers based on these online discussions?

C. Scope & Limitations

- 1) Considering the topic utilizes a Reddit discussion dataset, the study will not take into account the network structure and geological distribution of the data. Hence, it is possible that some sentiments or topics may come from foreigners or non-farmers.
- 2) The analysis is limited to text content and does not incorporate multimedia elements such as images or videos, which may also influence discussions or public perception.
- 3) Due to difficulties in data collection, the study will focus on Reddit and Rappler as the primary sources of data, which may not fully represent the entire spectrum of public opinion on Philippine agriculture.
- 4) The study will focus on general sentiments and topics, without delving into specific sub-sectors of agriculture (e.g., rice farming, livestock, fisheries).
- 5) The data collected is limited to: for Reddit, 2016 to 2025; for Rappler, 2023 to 2025.

TABLE I
RAW REDDIT DATA SAMPLE

post_title	Sample title about farming
post_url	https://reddit.com/...
comment	Agriculture in the Philippines needs reform.
comment_author	user123
comment_score	12
post_body	Full post discussing rice production and its challenges in the Philippines.
post_score	256
post_author	authorA
created_utc	2025-10-23 08:14:00

TABLE II
ONLINE NEWS PLATFORM DATA SAMPLE

title	Arms to farms: Maguindanao del...
link	https://www.rappler.com/philippines/...
date_published	2025-10-24 09:30:00
author	Maria Dela Cruz
text	COTABATO CITY, Philippines —Maguindanao del...

II. METHODOLOGY

A. Data Collection & Preprocessing

Reddit data were collected from r/Philippines filtered with ‘Agriculture’ via PRAW, while agriculture-related news articles were obtained from **Rappler**, specifically topics under ‘Department of Agriculture’. The metadata gathered from both sources are summarized in their respective tables (*see Table I and Table II*). Data cleaning procedures included: (i) removing null or missing values, (ii) removing special characters, (iii) normalizing text, and (iv) identifying and removing stopwords.

For Reddit, the final cleaned dataset includes a new column `post_full`, formed by concatenating `post_title` and `post_body`. Only the original posts were analyzed, as these contain the primary thematic content and clearly express the core issues being discussed, whereas comment sections introduce high levels of noise and irrelevant conversational chatter.

For Rappler, the cleaned dataset includes a new column `cleaned_text`, consisting of the concatenated and normalized article title and full article body.

B. Exploratory Data Analysis (EDA)

At this stage, basic exploratory data analysis (EDA) methods are implemented to provide a deeper understanding of what the datasets have.

1) *Basic Overview and Structure:* The Reddit dataset contains 12,730 entries with six columns, including post titles, URLs, comments, timestamps, and a combined `post_full` field. Most posts include comments, while only a small subset contain extended post bodies. The Rappler dataset consists of 121 news articles with five complete fields (`title`, `link`, `date_published`, `text`, and `cleaned_text`), all of which are non-null and uniformly structured.

2) *Word Frequency Analysis:* Word frequency analysis was performed using matplotlib’s WordCloud to visualize the most prominent terms in each corpus. Both the Reddit and Rappler word clouds exhibit similarities, as both prominently feature agriculture-related vocabulary. However, the Reddit word cloud (Fig. 1) also contains names of politicians and colloquial expressions such as “lalala pa” (*going to get worse*), “sadly undervalued,” and “brgy clearance,” which suggest negative sentiment or frustration commonly expressed in online discussions.

In contrast, the Rappler word cloud (Fig. 2) places stronger emphasis on rice-related issues. This aligns with reports from the Department of Agriculture acknowledging declining rice production due to several factors. Frequent mentions of terms such as “per kilo,” “rice price,” and “import” further highlight recurring concerns over agricultural commodity pricing, consistent with well-known challenges in the Philippine agricultural sector.

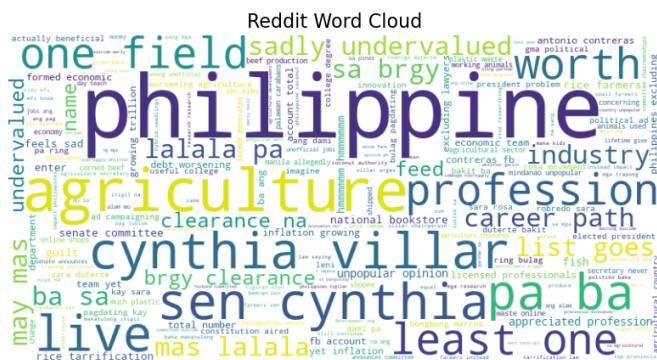


Fig. 1. Word frequency visualization for the Reddit agriculture corpus.

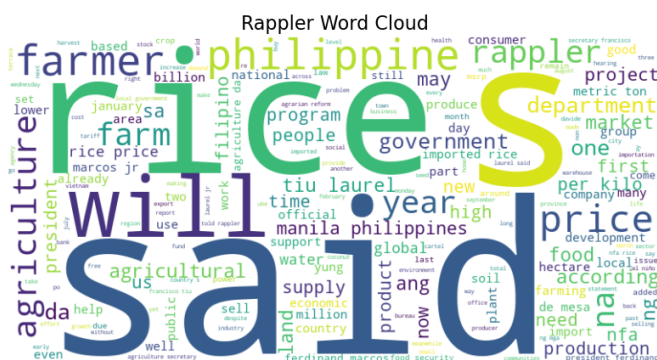


Fig. 2. Word frequency visualization for the Rappler agriculture corpus.

Additionally, it is noted that the top 20 most frequent Reddit terms include *agriculture*, *rice*, *agricultural*, *philippines*, and *profession*. For Rappler, the most frequent terms include *rice*, *agriculture*, *food*, and *farmers*, reflecting coverage focused on production, pricing, and food security.

3) *Post Frequency Over Time*: Understanding the distribution of posts across time provides context for identifying periods of heightened discourse within the study.

For Rappler, article publications appear sparser due to the relatively small dataset (approximately 120 articles). It is possible that a more comprehensive sample, potentially collected from multiple news outlets, would show clearer temporal patterns in agricultural reporting.

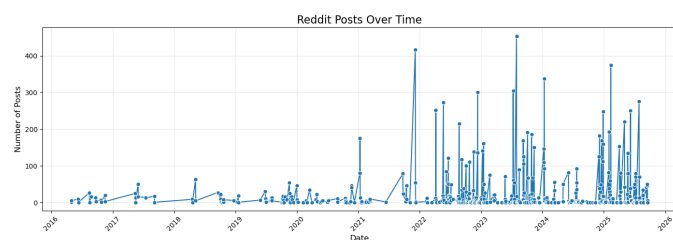


Fig. 3. Post frequency over time for the Reddit agriculture corpus.

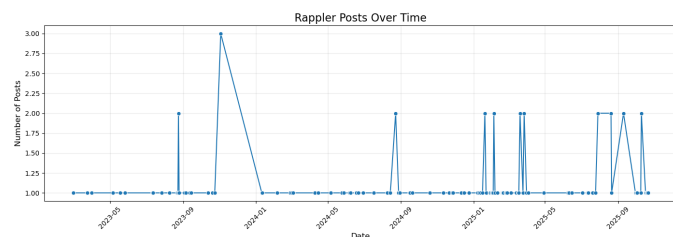


Fig. 4. Article publication frequency over time for the Rappler agriculture corpus.

News articles similarly emphasize *prices*, *farmers*, and *agricultural production*, indicating persistent concerns about economic pressures and commodity pricing. Given the prominence of these terms in both Reddit discourse and Rappler reporting, it is evident that issues surrounding rice supply, pricing, and farmer livelihood remain central topics of public conversation.

In this stage, several NLP methods are applied to derive deeper insights from the textual data. First, a Bag-of-Words (BoW) word cloud is generated to provide a cleaner and less noisy visualization of prominent terms compared to the basic word-frequency word cloud used in the EDA stage. Next, topic modeling is employed to uncover the major latent themes present in both datasets. Finally, sentiment analysis is conducted to assess the general polarity of the discourse and evaluate how positively or negatively these topics are framed.

Additionally, the results of the Reddit BoW (Fig. 5) and Rappler BoW (Fig. 6) further reinforce that discussions on agriculture consistently revolve around key themes such as *government, policies, prices, undervalued, the Department of Agriculture (DA), and food*. Even with a more semantically filtered and less noisy representation, the dominant terms remain strongly aligned, indicating that both public discourse and news reporting emphasize similar core issues within the agricultural sector. Additionally, the BoW word clouds [Fig. 6, Fig. 5](*i.e., agriculture, undervalued, department, clearance etc.*) highlights finer or more specific terms compared to the EDA word clouds [Fig. 2, Fig. 1](*rice, said, will, philippine, farmer, etc.*) which contained more noisy and general terms when it comes to agricultural concerns.

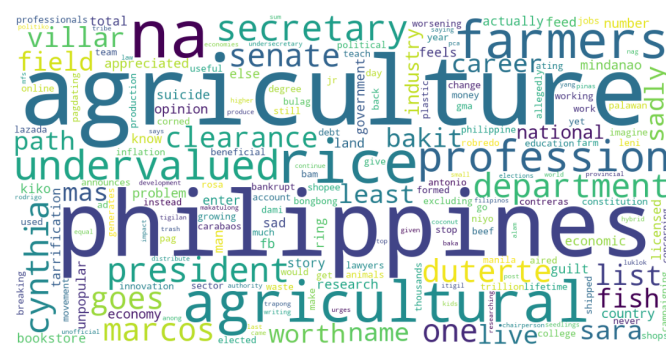


Fig. 5. Reddit word cloud based on Bag-of-Words modeling.

2) *Topic Modelling*: Topic modeling was conducted using a Bag-of-Words representation and Latent Dirichlet Allocation (LDA) to uncover the latent themes present in the Reddit and Rappler corpora. LDA groups together words that frequently co-occur across documents, enabling the identification of coherent thematic clusters. For this study, five topics were generated for each dataset. Tables III and IV present the top



Fig. 6. Rappler word cloud based on Bag-of-Words modeling.

five representative words for every topic, along with their corresponding interpretations.

TABLE III
LDA TOPICS FOR REDDIT (TOP WORDS AND INTERPRETATION)

T	Top Words	Interpretation
1	africa, cases, fever, importing, million	Animal disease outbreaks, importation issues, and government announcements.
2	hectare, financial, living, businesses, importations	Farming operations, livelihood concerns, and financial pressures.
3	heat, irrigation, growing, distributed, energy	Climate impacts, irrigation infrastructure, and support distribution.
4	affordable, harvest, records, millions, group	Harvest output, affordability, and community-level food concerns.
5	customs, association, establish, promote, quimbo	Trade regulation, customs processes, and institutional actors.

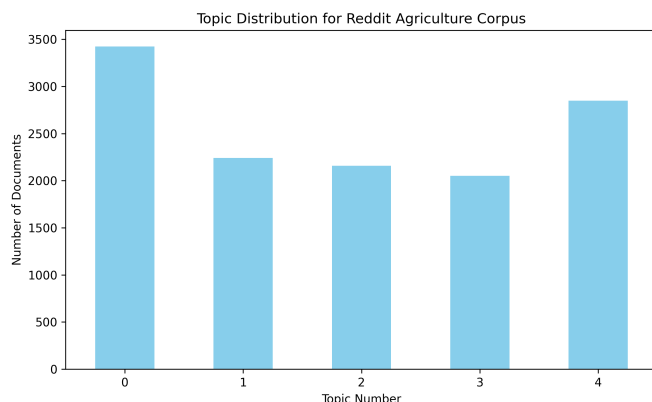


Fig. 7. Distribution of LDA-derived topics for the Reddit agriculture corpus.

3) *Sentiment Analysis:* Sentiment analysis was conducted to examine the underlying emotional polarity present in both datasets. The tabularisai/multilingual-sentiment-analysis model [4], a BERT-based multilingual classifier capable of handling both English and Filipino text, was employed to evaluate each Reddit post and Rappler article. This model assigns one of five sentiment categories: *Very Negative*,

TABLE IV
LDA TOPICS FOR RAPPLER (TOP WORDS AND INTERPRETATION)

T	Top Words	Interpretation
1	farmers, farm, rice, food, soil	Farming practices, soil conditions, and crop production.
2	price, rice, prices, supply, government	Rice pricing, supply fluctuations, and economic pressures.
3	agriculture, land, reform, agrarian, marcos	Agrarian reform, land issues, and sector governance.
4	rice, da, nfa, kilo, imported	Importation, retail pricing, and regulatory agency functions.
5	agriculture, irrigation, production, project, national	Government programs, irrigation efforts, and national production initiatives.

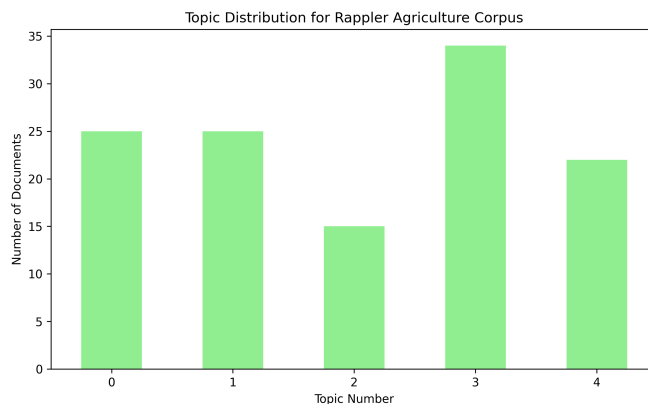


Fig. 8. Distribution of LDA-derived topics for the Rappler agriculture corpus.

Negative, Neutral, Positive, and Very Positive, along with a confidence score (*Default labels by tabularisai*).

Tables V and VI summarize the distribution of sentiment categories for each dataset. Reddit displays a wider spread of sentiments, with notable proportions of negative and very negative posts, reflecting public frustrations related to pricing, farmer livelihood, government policies, and supply issues. Rappler, as a journalistic source, maintains predominantly neutral sentiment across topics, consistent with its information-oriented reporting style.

TABLE V
SENTIMENT DISTRIBUTION FOR THE REDDIT CORPUS

Sentiment	Count
Neutral	8368
Negative	2802
Very Negative	1098
Positive	350
Very Positive	112

TABLE VI
SENTIMENT DISTRIBUTION FOR THE RAPPLER CORPUS

Sentiment	Count
Neutral	113
Negative	6
Very Positive	2

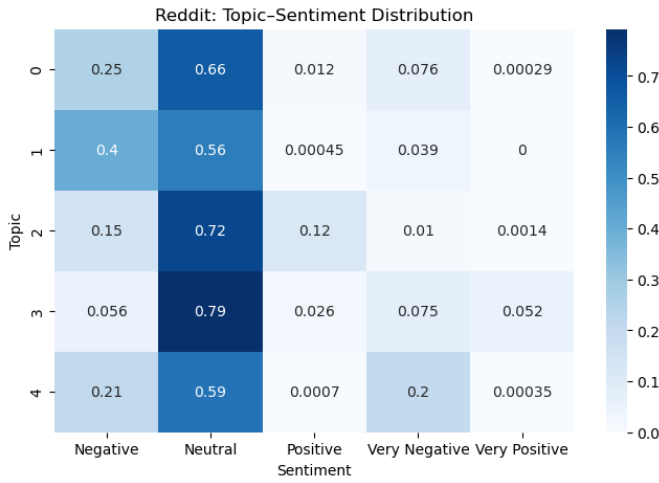


Fig. 9. Reddit Topic-Sentiment Heatmap Distribution

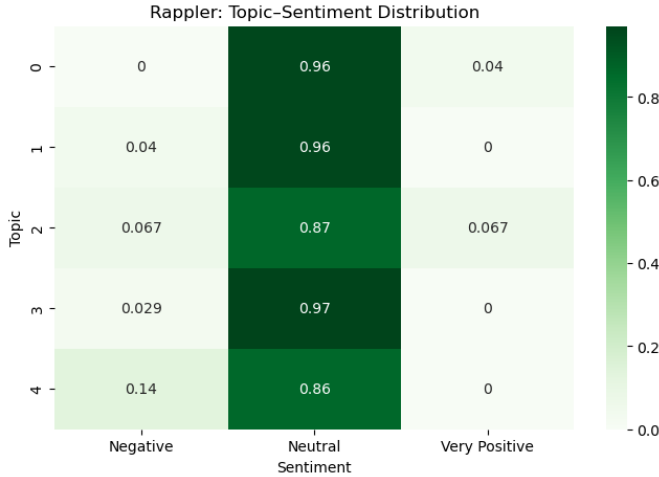


Fig. 10. Rappler Topic-Sentiment Heatmap Distribution

III. RESULTS & DISCUSSION

The Reddit and Rappler datasets reveal distinct and overlapping themes related to the challenges of the Philippine agricultural sector. Across both platforms, the results consistently highlight public attention toward issues involving pricing, undervaluation of the sector, and the persistent struggles faced by farmers.

The **exploratory data analysis (EDA)** provides the foundation for these observations. Word frequency patterns show that both datasets emphasize terms associated with agricultural commodity prices, policy discussions, and expressions of concern or frustration, such as *“lalala pa”* (going to get worse). Mentions of political actors also appear frequently, reflecting the strong relationship between governance and agricultural outcomes.

Topic modeling results from the Reddit corpus reveal several prominent themes, including animal disease outbreaks (such as African swine fever), importation and pricing poli-

cies, and broader socioeconomic concerns affecting farmers. These topics align with ongoing national issues where farmers struggle to compete against imported agricultural goods and face volatile market conditions.

In the Rappler corpus, topic modeling captures thematically similar but more structured discussions. Since the dataset is derived from Rappler’s coverage of the Department of Agriculture, the topics reflect journalistic reporting on farming practices, crop production, price movements, importation activities, and government policy initiatives. Compared to Reddit, the framing is more formal and informational, consistent with news media practices.

Sentiment analysis revealed that both datasets are dominated by neutral sentiment (see Tables V and VI). Given that Rappler primarily presents informational reporting, its predominantly neutral and unbiased tone is expected. Reddit, on the other hand, exhibits a noticeably higher concentration of negative sentiment. The strongest negativity appears in Topic 2, which relates to farming operations, livelihood concerns, and financial pressures, followed by Topic 1, which covers animal disease outbreaks, importation issues, and government announcements.

Considering these results, public sentiment aligns with the Department of Agriculture’s own announcements regarding the major challenges faced by the country’s agricultural sector [1].

IV. CONCLUSION

This study demonstrates the alignment between public discourse on Reddit and journalistic coverage from Rappler regarding agricultural issues in the Philippines. The recurring topics identified across both datasets such as pricing, importation, production challenges, and farmer livelihood, reflect widely acknowledged national concerns. Although negative sentiment was less prominent than expected, this may be attributed to the nature of the collected Reddit data, which focused only on `post_title` and `post_body`, both of which tend to contain more informational rather than emotionally charged content. The absence of `post_comment` data may have limited the detection of stronger opinions or polarized expressions from users.

Based on the findings of the study, it is evident that both public discussions and news reporting have some discrepancies. While most topics are aligned, there are still some topics that are not as covered in news articles such as animal diseases, irrigation, and economical pressures of farmers. This may be due to the fact that news articles tend to focus on more formal and structured topics, especially concerned with government policies presented by the Department of Agriculture.

This study highlights the importance of understanding public sentiment and discourse surrounding agriculture in the Philippines. By analyzing both Reddit discussions and Rappler articles, we gain insights into the challenges faced by farmers and the broader agricultural sector. These insights can inform policymakers, stakeholders, and the public about the pressing issues that need to be addressed to support and improve the agricultural landscape in the country.

Additionally, the study acknowledges the fact that the data collected and analyzed may not fully represent the entire spectrum of farmers' needs and concerns. This is because the data is limited to online discussions, where authors may or may not necessarily be farmers themselves. Other concerns such as geographical scenario (are these actual farmers? researchers? agricultural enthusiasts?). Language barrier may also be present in the study considering Reddit data may include tagalog (or other dialects) that may not necessarily be captured by the models used in the study.

It is recommended that further research on the topic be conducted with a more domain-rich dataset, considering the low-resource nature of the topic (Abundant resources on government policies, but limited resources on farmers' needs and concerns).

V. POSSIBLE FUTURE WORKS

Future research may build upon this study by incorporating more domain-rich and diverse datasets to deepen the analysis. Potential directions include:

- 1) Expanding to larger and more varied social media platforms, such as Facebook or X. Additionally, data directly from interviews or farmer testimonials or farmer cooperatives could provide richer, firsthand perspectives. This is to ensure that the analysis captures a broader spectrum of public discourse and farmer experiences.
- 2) Integrating institutional documents and reports from agricultural agencies such as the Department of Agriculture (DA), Department of Agrarian Reform (DAR), and PhilRice. Similar to the News articles, these can provide relevant documents to compare and ground the public discourse analysis with facts. (*Similar to Retrieval-Augmented Generation (RAG) approach*)
- 3) Implement more advanced NLP techniques, such as transformer-based models to encode-decode agricultural text data, and predict through generative AI such as Ollama/Sea-lion/GPT models possible solutions to the challenges faced by the agricultural sector, based on the knowledge base.

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